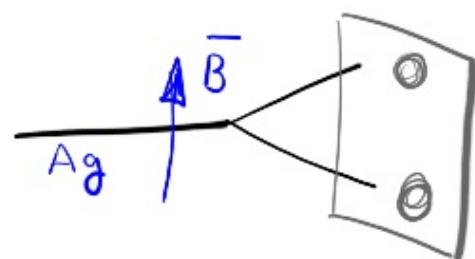
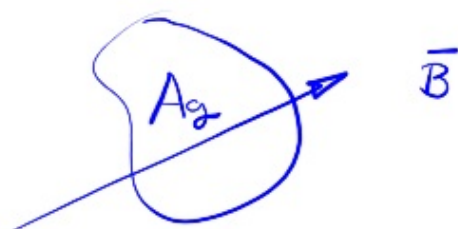


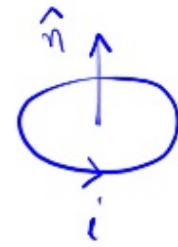
Spín y momento magnético

1) $\lambda = 589,3 \text{ nm}$ (sodio) $\begin{matrix} < 589,0 \text{ nm} \\ < 589,6 \text{ nm} \end{matrix}$

2)



Spín



$$\bar{\mu} = \gamma A \hat{n}$$

$$\bar{\mu} = \frac{q}{2mc} \bar{L}$$

$$\underset{\substack{\uparrow \\ \text{momento} \\ \text{magnético}}}{\bar{\mu}_e} = -g \frac{|e|}{2m_e c} \underset{\substack{\uparrow \\ \text{SPIN (momentum angular intrínseco)}}}{\hat{S}}$$

Magnetón de Bohr

$$M_B = \frac{|e| \hbar}{2m_e c}$$

$$\vec{\mu}_e = -g M_B \hat{S}$$

↑
razón giromagnética $\approx 2,0022$

Al igual que el electrón, los protones y neutrones manifiestan momento magnético.

$$\vec{\mu} = g \frac{|e| \hbar}{2M c} \hat{S}$$

↑
magnetón nuclear

$$g_p = 5,59$$

$$g_n = -3,85$$

neutrón udd (quarks)

$$\frac{2}{3}e \quad -\frac{e}{3} \quad -\frac{e}{3}$$

Contribución cuadrupolar

$$\hat{S} = \frac{1}{2} \hbar \sigma \leftarrow \text{operador}$$

$$\sigma = (\sigma_x, \sigma_y, \sigma_z)$$

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

matrices de Pauli

$$\sigma_j^+ = \sigma_j$$

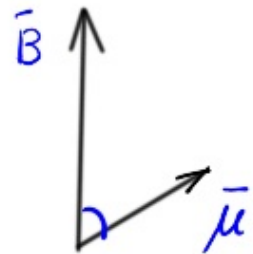
$$\sigma_i \sigma_j = \delta_{ij} + i \epsilon_{ijk} \sigma_k$$

$$[\sigma_i, \sigma_j] = 2i \epsilon_{ijk} \sigma_k$$

$$\sigma \cdot \sigma = \sigma^2 = 3 \cdot \mathbb{1}$$

El momentum angular satisface

$$[\hat{S}_i, \hat{S}_j] = i \hbar \epsilon_{ijk} \hat{S}_k$$



$$H = - \overset{\text{Operator}}{\mu_e} \cdot \overset{\text{Operator}}{B}$$

$$\mu_e = -g M_B \hat{S}, \quad M_B = \frac{|e| \hbar}{2 m_e c}$$

Considerando $B = B_0 \hat{k}$

$$\begin{aligned} \rightarrow H &= \frac{1}{2} g M_B \hbar \hat{\sigma} \cdot B_0 \hat{k} \\ &= \underbrace{\frac{1}{2} g M_0 \hbar B_0}_{E_0} \sigma_z = E_0 \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \end{aligned}$$

Resolventen $\hat{H} |\phi\rangle = i\hbar \partial_t |\phi\rangle$
 $\uparrow$
 espinor

$$E_0 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \phi_1 \\ \phi_2 \end{pmatrix} = i\hbar \partial_t \begin{pmatrix} \phi_1 \\ \phi_2 \end{pmatrix}$$

$$\begin{aligned} \Rightarrow E_0 \phi_1 &= i\hbar \partial_t \phi_1 \rightarrow \phi_1 = c_1 e^{-i \frac{E_0}{\hbar} t} \\ -E_0 \phi_2 &= i\hbar \partial_t \phi_2 \rightarrow \phi_2 = c_2 e^{i \frac{E_0}{\hbar} t} \end{aligned}$$

$$|\psi_t\rangle = c_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} e^{-i\omega t} + c_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix} e^{i\omega t}, \quad E_0 = \hbar\omega$$

$$\langle \psi_t | = c_1^* (1 \ 0) e^{i\omega t} + c_2^* (0 \ 1) e^{-i\omega t}$$

$$\langle \psi_t | \sigma_z | \psi_t \rangle = |c_1|^2 - |c_2|^2$$

$$\langle \psi_t | S_z | \psi_t \rangle = \frac{\hbar}{2} (|c_1|^2 - |c_2|^2)$$

$$\langle \psi_t | S_x | \psi_t \rangle = \frac{\hbar}{2} (c_1^* c_2 e^{2i\omega t} + c_1 c_2^* e^{-2i\omega t})$$

$$\langle \psi_t | S_y | \psi_t \rangle = \frac{\hbar}{2} (i c_1 c_2^* e^{-2i\omega t} - i c_1^* c_2 e^{2i\omega t})$$

Supongamos que en $t=0$, la orientación del SPIN
se alinea completamente según el eje x

$$\hat{S}_x |\psi\rangle \Big|_{t=0} = \frac{\hbar}{2} |\psi\rangle \Big|_{t=0}$$

$$\hat{S}_x |\psi\rangle = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = \frac{\hbar}{2} \begin{pmatrix} c_2 \\ c_1 \end{pmatrix} \rightarrow$$

$$\begin{pmatrix} c_2 \\ c_1 \end{pmatrix} = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} \rightarrow c_1 = c_2 \wedge |c_1|^2 + |c_2|^2 = 1$$

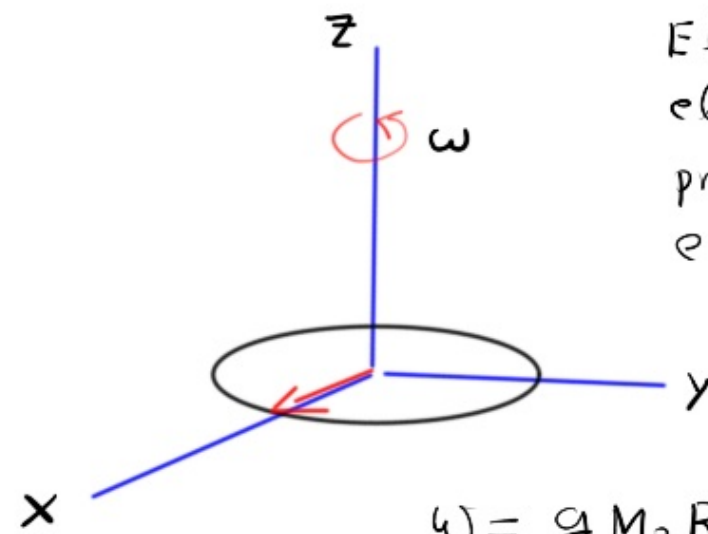
$$\rightarrow |c_1| = |c_2| = \frac{1}{\sqrt{2}}$$

$$\text{consideremos reales : } c_1 = c_2 = \frac{1}{\sqrt{2}}$$

$$\langle \hat{S}_z \rangle_\psi = 0$$

$$\langle \hat{S}_y \rangle_\psi = \frac{1}{2} \hbar \sin 2\omega t$$

$$\langle \hat{S}_x \rangle_\psi = \frac{1}{2} \hbar \cos 2\omega t$$



El spin precesa en el plano xy con una proyección nula según el eje z .

$\omega = g M_B B_0$: frecuencia de precesión